

Diary entry for Spectral-Sequences

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1 Theory

2 Practice

Courtesy of Ajay.

Exercise 2.1. Use the Atiyah-Hirzebruch spectral sequence to compute the K -theory of $\mathbb{C}P^\infty$.

Exercise 2.2. Compute the homology of $K(\mathbb{Z}, 2) = \mathbb{C}P^\infty$ using the Serre spectral sequence, noting that there is a fibration $S^\infty \rightarrow \mathbb{C}P^\infty$ (what is its fiber?). Note that you should be able to verify your answer easily using cellular homology.

Exercise 2.3. Do the same as the above exercise for $\mathbb{R}P^\infty$ (perhaps with $\mathbb{Z}_2$ coefficients; I am not sure how tractable this is).

Exercise 2.4. If $E \rightarrow B$ is a covering space, how are the homology of E and B related?

Exercise 2.5. Deduce the Künneth formula for (co)homology with coefficients in a field using the Serre spectral sequence.

Bonus question: What about the case where the coefficient ring is not a field?

Exercise 2.6. Compute the cohomology of ΩS^n , the loop space of the n -sphere.

Exercise 2.7. If $E \rightarrow B$ is a fibration whose fiber is contractible, what can be said about the homology of E and B ? (For a spoiler of the answer, feed the fibration into the homotopy long exact sequence.) What about if B is contractible, or E is contractible?